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ROLL NO:-73

BRANCH :-CSE(IOT)

LAB 07	
Aim:	Extraction of data from files and user in R : Import the data from different file formats to R. Export data from R to different file formats. Apply basic functions to extract data.
Theory	In R, data can be imported from and exported to various file formats, allowing users to work with different types of data. Here's an overview of how this works: **Importing Data into R** R provides multiple functions to read data from various file formats, such as: 1. **CSV Files**: Comma-separated value (CSV) files are a common format for tabular data. R can easily import CSV files into data frames, making it suitable for further analysis and manipulation. 2. **Excel Files**: Excel spreadsheets are widely used in business and research. R can import data from Excel files using specialized libraries designed for reading .xls or .xlsx formats. 3. **Text Files**: R can handle plain text files, including tab-delimited or space-delimited files, allowing users to read raw textual data into R. 4. **RDS Files**: R has its own native format for saving data objects, called .rds. This format allows users to save R objects in a binary format, which can be read back into R at a later time. 5. **JSON and Other Formats**: JSON is another widely used data format for representing structured data. R can read and process JSON files, as well as other formats like XML, using dedicated packages. **Exporting Data from R** Once data is processed and analyzed in R, users may need to export the results into various formats for sharing, reporting, or further analysis. R supports exporting data to formats such as: 1. **CSV Files**: R can export data frames to CSV files, making it easy to share datasets in a format that is widely compatible with other software, like Excel or Google Sheets. 2. **Excel Files**: R can export data to Excel files, allowing users to save their results in a familiar spreadsheet format, which can be easily opened and manipulated in Excel. 3. **RDS Files**: R objects can be saved in the .rds format, enabling users to preserve and reload entire data objects, including complex structures like models and lists. 4. **Text Files**: R can write data to text files, including space- or tab-delimited formats, which is useful for sharing data in a plain-text format that can be read by other software. 5. **JSON Format**: R can export data to JSON format, which is commonly used for web services and APIs to exchange data in a structured manner. **Basic Functions for Data Manipulation** Once data is imported into R, a variety of functions can be used to extract and manipulate the data. These include basic operations like viewing the structure of data, summarizing data, selecting subsets of data, or applying more complex transformations. • **Summarizing Data**: Functions like summary() provide a summary of key statistics for

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	each variable in the dataset. • **Viewing Data**: Functions such as head() and tail() allow users to preview the first and last few rows of a dataset. • **Selecting Data**: Functions like subset() help in selecting specific rows or columns based on certain conditions.
Program:	**i) Main code** #Sheikh Wasimuddin 23071044 IOT-73 library(readxl) library(haven) library(dplyr) library(writexl) data_csv <- read.csv("yourfile.csv") write.csv(data_csv, "output_file.csv", row.names = FALSE) write_xlsx(data_csv, "output_file.xlsx") write.table(data_csv, "output_file.txt", sep = "\t", row.names = FALSE, quote = FALSE) saveRDS(data_csv, "output_file.rds") print(data_csv[, c("age", "salary")]) # Subset columns print(subset(data_csv, age > 30)) # Filter using subset print(filter(data_csv, age > 30)) # Filter using dplyr summary(data_csv) mean_value <- mean(data_csv$age, na.rm = TRUE) print(mean_value) print(data_csv %>% group_by(salary) %>% summarize(Mean_Age = mean(age, na.rm = TRUE))) print(apply(data_csv[, c("age", "salary")], 2, sum, na.rm = TRUE)) print(apply(data_csv[, c("age", "salary")], 1, sum, na.rm = TRUE)) print(lapply(data_csv[, c("age", "salary")], mean, na.rm = TRUE))

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	ii) CSV File code
	#Sheikh Wasimuddin 23071044 IOT-73
	data <- data.frame(
	  id = 1:5,
	  name = c("Tanmay", "Teju", "Wasim", "Vedant", "saurant"),
	  age = c(20, 19, 20, 22, 30),
	  salary = c(5, 4, 3, 2, 1)
	)
	cat("Here is the sample data:\n")
	print(data)
	write.csv(data, "yourfile.csv", row.names = FALSE)
	cat("\nThe data has been saved as 'yourfile.csv'.\n")

Output:	

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```r
library(readxl)
library(haven)
library(dplyr)
library(writexl)
data_csv <- read.csv("yourfile.csv")
write.csv(data_csv, "output_file.csv", row.names = FALSE)
write_xlsx(data_csv, "output_file.xlsx")
write.table(data_csv, "output_file.txt", sep = "\t", row.names = FALSE, quote = FALSE)
saveRDS(data_csv, "output_file.rds")
print(data_csv[, c("age", "salary")])  # Subset columns
print(subset(data_csv, age > 30))  # Filter using subset
print(filter(data_csv, age > 30))  # Filter using dplyr
summary(data_csv)
mean_value <- mean(data_csv$age, na.rm = TRUE)
print(mean_value)
print(data_csv %>% group_by(salary) %>% summarize(Mean_Age = mean(age, na.rm = TRUE)))
print(apply(data_csv[, c("age", "salary")], 2, sum, na.rm = TRUE))
print(apply(data_csv[, c("age", "salary")], 1, sum, na.rm = TRUE))
print(lapply(data_csv[, c("age", "salary")], mean, na.rm = TRUE))
```

2:16 (Top Level) ÷

Console Terminal Background Jobs

R 4.4.0 · ~/

```
> source("~/.active-rstudio-document")
  age salary
1  20      5
2  19      4
3  20      3
4  22      2
5  30      1
[1] id     name   age    salary
<0 rows> (or 0-length row.names)
[1] id     name   age    salary
<0 rows> (or 0-length row.names)
[1] 22.2
# A tibble: 5 × 2
  salary Mean_Age
   <int>    <dbl>
1      1       30
2      2       22
3      3       20
4      4       19
5      5       20
   age salary
   111     15
[1] 25 23 23 24 31
$age
[1] 22.2

$salary
[1] 3

>
```

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```r
data <- data.frame(
  id = 1:5,
  name = c("Tanmay", "Teju", "Wasim", "Vedant", "saurant"),
  age = c(20, 19, 20, 22, 30),
  salary = c(5, 4, 3, 2, 1)
)
cat("Here is the sample data:\n")
print(data)
write.csv(data, "yourfile.csv", row.names = FALSE)
cat("\nThe data has been saved as 'yourfile.csv'.\n")

-
```

12:2 (Top Level)

Console Terminal Background Jobs

R 4.4.2 · ~/

```
[1] 3

>
>
>
> source("~/.active-rstudio-document")
Here is the sample data:
  id    name age salary
1  1  Tanmay  20      5
2  2    Teju  19      4
3  3   Wasim  20      3
4  4  Vedant  22      2
5  5 saurant  30      1

The data has been saved as 'yourfile.csv'.
>
```

Conclusion	Extraction of data from files and user in R : Import the data from different file formats to R. Export data from R to different file formats. Apply basic functions to extract data executed successfully.
Name Of Faculty:	Prof. Akhil Jajulwar